# MAIN ARTICLE REMOVED CONTENT | page 1

Abstract DFT/pathway conclusion from beta-O-4 model compound.

while preserving cellulose intact. Reaction pathway studies and density functional theory calculations based on a β-O-4 model compound reveal three coexisting pathways. The $Pt_1Ni$ alloy preferentially promotes the dehydroxylation of $C_\alpha$-OH and forms a key $C_\alpha = C_\beta$ intermediate due to the oxygen affinity of Ni sites, and ultimately, enhance the production of valuable propenyl products via the synergistic effect of Pt and Ni. This work provides a strategy

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Intro paragraph stating DFT calculations reveal three pathways and barriers.

stability in recyclability experiments. Model compound catalytic experiments and density functional theory (DFT) calculations reveal that three pathways including $C_\alpha$-OH deH, $C_\alpha$-OH deOH-I, and $C_\alpha$-OH deOH-II co-existed in lignin depolymerization. $Pt_1Ni$ SAA catalyst could reduce the energy barrier for dehydroxylation, rendering it to out-perform pure Pt catalyst in $C_\alpha$-OH deOH. Moreover, $C_\alpha$-OH deOH is superior to the $C_\alpha$-OH deH pathway, thereby promoting the selectivity towards useful propenyl-ended products. During the preparation of

## DFT calculations

To gain a deeper understanding of the reaction mechanism and pathways, the DFT calculations were performed to investigate the mechanism of SG hydrogenation on the exposed (100) facet of Pt and $Pt_1Ni$. Figure S17 shows the SG could preferentially chemisorb on the $Pt_1Ni(100)$ and Pt(100) surfaces via the flat-lying configuration, at which both benzene rings in SG were bonded to the surfaces. The higher binding strength between benzene rings and Pt(100) indicates the stronger carbophilicity of platinum. The conversion of chemisorbed SG can potentially start with the dehydrogenation of $(C_\alpha)OH$, the hydrogenolysis of $C_\alpha$-OH, the ether C-O bond cleavage, or the dehydrogenation of $C_\alpha$-H. Among these steps, the lowest free energy barrier step is the hydrogenolysis of $C_\alpha$-OH over $Pt_1Ni(100)$, while it is the dehydrogenation of $(C_\alpha)OH$ over Pt(100) (Fig. S18). This indicates that the hydrogenolysis of $C_\alpha$-OH would preferentially occur over $Pt_1Ni(100)$, while the dehydrogenation of $(C_\alpha)OH$ would be more favorable over Pt(100). This could be attributed to the higher oxophilicity of nickel compared to platinum[44]. The Crystal Orbital Hamilton Populations (COHP) analysis (Fig. S19) confirms that the strong adsorption of oxygen atoms by Ni sites on the $Pt_1Ni$ (100) surface weakens the $C_\alpha$-OH bond compared to Pt(100), resulting in a 0.04 Å elongation and thereby facilitating $C_\alpha$-OH bond cleavage. It indicates that $Pt_1Ni$ (100) might be more favorable to produce the intermediates with unsaturated C = C bond, such as **4a**, while Pt (100) is more favorable to produce the intermediates with carbonyl group, such as **2a**, at the initial stage of the SG hydrogenation. This aligns with the observed differences in product distribution (Table 2) and the reaction pathways proposed based on experimental observations.

To distinguish how these initial steps govern subsequent reactivity, we conducted a further mechanistic study of the reaction pathways, commencing with the hydrogenolysis of $C_\alpha$-OH and the dehydrogenation of $C_\alpha$-OH over the two catalysts. The corresponding pathways were labeled as $C_\alpha$-OH deOH towards the production of **8a, 6a**, and **11a**, and $C_\alpha$-OH deH towards the production of **7a**, respectively. In the $C_\alpha$-OH deH pathway (Fig. S20), each elementary step exhibits a higher energy barrier on $Pt_1Ni$ (100) than Pt (100). Specifically, the water-mediated dehydrogenation of the $C_\alpha$-H for the production of **2a** encounters a free energy barrier of 1.11 eV (TS2-1) on $Pt_1Ni(100)$, while the subsequent ether C-O bond cleavage requires an even higher barrier of 1.51 eV (TS3-1). In contrast, the higher carbophilicity of Pt sites lead to the reduced energy barriers for the dehydrogenation of $C_\alpha$-H (TS2-1′, 0.77 eV) and the ether C-O bond breaking (TS3-1′, 0.80 eV) over Pt(100). These results imply higher selectivity towards **7a** over Pt(100) compared to $Pt_1Ni(100)$, which aligns well with the product distribution listed in Table 2. The $C_\alpha$-OH deOH branches into two pathways after the dehydroxylation: (I) hydrogenation of $C_\alpha$-H to produce **3a** first and (II) dehydrogenation of $C_\beta$-H to produce **4a** first. In the $C_\alpha$-OH deOH-I pathway (Fig. S21), the least stable TS was found in the hydrogenolysis of $C_\alpha$-OH (TS1-2′) on Pt(100), which is less stable than the rate-determining TS of the ester C-O bond breaking (TS3-2, 0.73 eV) on $Pt_1Ni$ (100). This would limit the **8a** production on Pt(100). Figure S22 shows that the energy barriers of most steps on $Pt_1Ni$(100) are lower than those on Pt(100) in the $C_\alpha$-OH deOH-II pathway. The TS of the hydrogenolysis of $C_\alpha$-OH on Pt(100) (TS1-2′) is still the most unstable TS, which is less stable than the rate-determining TS for the ester C-O bond breaking on $Pt_1Ni$(100) (TS3-3). Moreover, the barrier of the ester C-O bond breaking on $Pt_1Ni$(100) ($G_a = 1.43$ eV) is lower that on Pt(100) ($G_a = 1.46$ eV). These suggest that $Pt_1Ni$(100) would surpass Pt(100) in the $C_\alpha$-OH deOH-I and $C_\alpha$-OH deOH-II pathways, while Pt(100) would be superior to $Pt_1Ni$(100) in the $C_\alpha$-OH deH pathway. Hence, the oxophilicity of nickel could facilitate the production of propenyl side-chained compounds on $Pt_1Ni$(100).

In Fig. 4, we energetically compared the three pathways to understand the selectivity of SG hydrogenation on the $Pt_1Ni$(100) surface. Notably, the rate-determining step in each pathway is the ether C-O bond cleavage. It needs to overcome the free energy barrier of 1.51 eV in $C_\alpha$-OH deH pathway, corresponding to the step from IM2-1 to IM3-1. Similarly, the ether C-O bond breaking step of IM2-2 to IM3-2 is the most difficult step in the $C_\alpha$-OH deOH-I pathway, requiring an energy input of 1.48 eV. In the $C_\alpha$-OH deOH-II pathway, the ether C-O bond breaking of IM2-3 to IM3-3 requires the lowest energy input of 1.43 eV among the three pathways. The lower energy barrier of the rate-determining C-O ether bond breaking indicates that the $C_\alpha$-OH deOH-II would be superior to the $C_\alpha$-OH deOH-I and $C_\alpha$-OH deH pathways, indicating a higher yield of **6a** and **11a** for SG hydrogenation. It is also in good agreement with our experimental observation. Noteworthily, the **6a** produced in $C_\alpha$-OH deOH-II pathway at the metal $Pt_1Ni$ site is an enol ether intermediate, which is a crucial intermediate in lignin depolymerization. The $C_\beta$-O bond of this intermediate could be further cleaved to produce propenyl-ended products.

Based on the results of model compound catalytic experiments and DFT calculations, we provide a possible mechanism of lignin depolymerization in this work (Fig. 5). Three pathways ($C_\alpha$-OH deH, $C_\alpha$-OH deOH-I, and $C_\alpha$-OH deOH-II) co-existed in lignin depolymerization.

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Discussion paragraph invoking density functional theory calculations.

recyclability under operational conditions. Model compound catalytic experiments and density functional theory (DFT) calculations reveal the coexistence of three pathways ($C_\alpha$-OH deH, $C_\alpha$-OH deOH-I, and $C_\alpha$-OH deOH-II) based on the different dimeric intermediates in lignin depolymerization and $Pt_1Ni$ SAA catalyst is superior to pure Pt catalyst for the Cα-OH deOH-I and Cα-OH deOH-II, while Pt(100) would be

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Discussion continuation about Pt1Ni(100)/Pt(100 pathway comparison.

superior to $Pt_1Ni(100)$ for the Cα-OH deH. Over $Pt_1Ni$ SAA catalyst, the $C_\alpha$-OH deOH II is easier due to the oxygen affinity of Ni sites and forms $C_\alpha = C_\beta$ double, leading to propenyl products formation through enol ether intermediate. In addition, compared to pure Pt catalyst, $Pt_1Ni$ SAA catalyst can lower the energy barrier for breaking C-O bonds, which is rate-determining step in three pathways. This work provides significant advances in the design of high-performance single-atom alloy catalysts for sustainable lignin depolymerization and contributes to the holistic valorization of lignocellulosic biomass in the context of biorefinery.

Figure 4 comparative DFT reaction pathway diagram and caption.

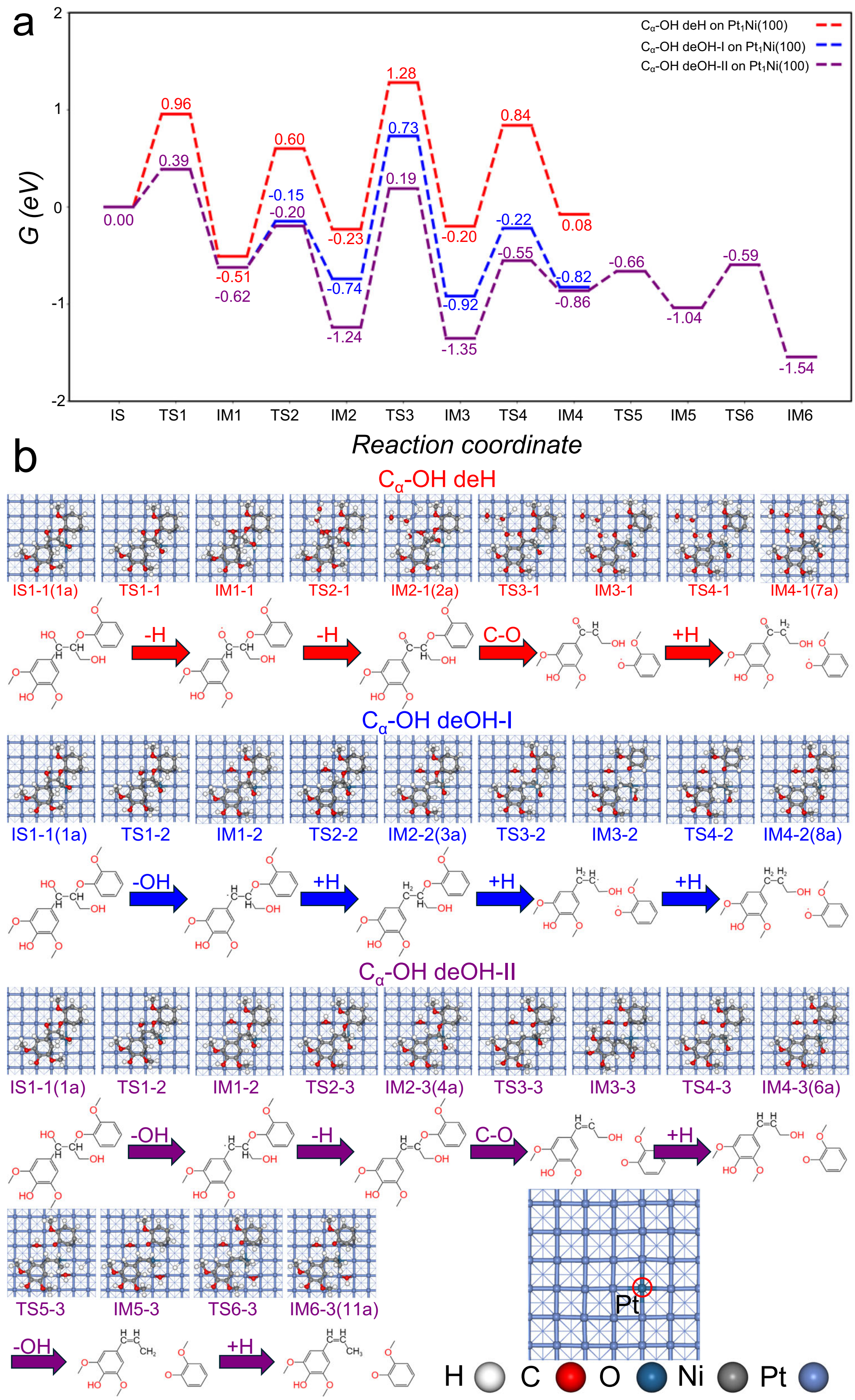


**Fig. 4 | The comparative diagram of the reaction pathways. a** Energy profiles for $C_\alpha$-OH deH, $C_\alpha$-OH deOH-I, and $C_\alpha$-OH deOH-II on $Pt_1Ni(100)$. **b** Corresponding the molecular structure diagram. The Pt atoms in $Pt_1Ni(100)$ have been circled in red in the last figure.

Computational details: DFT method setup.

## Computational details

All the calculations were performed in the framework of periodic density functional theory (DFT) using the Perdew-Burke-Ernzerhof (PBE) functional[48] implemented in Vienna ab initio simulation package (VASP 6.2.0)[49–51]. The electron-ion interactions were modeled utilizing the projector augmented wave (PAW) pseudopotential and the plane wave basis-set was expanded to a converged cut-off energy of 450 eV. The energy calculation was performed using Methfessel-Paxton smearing method with the width of 0.05 eV. The van der Waals interaction was considered by the empirical DFT-D3(BJ) method[52,53]. The

Computational details continuation: slabs, energies, TS and COHP analysis.

adsorption energy ($E_{ads}$) was calculated by the following equation:

$$E_{ads} = E_{X^*} - E_{surf} - E_X \quad (3)$$

where $E_{X^*}$, $E_{surf}$, and $E_X$ are the total energies of the adsorbate on the surface, bare surface, and gaseous adsorbate molecule. The asterisk represents the adsorbate on the surface.

The Gibbs free energy of adsorption ($G_{ads}$) was calculated by the $E_{ads}$ with the thermodynamic correction under the reaction condition of 413.15 K as follows:

$$G_{ads} = E_{ads} + \Delta ZPE + \Delta(\Delta U) - T\Delta S^{\Theta} + RT + RT\ln\left(\frac{p}{p^{\Theta}}\right) + RT\ln(x_i) \quad (4)$$

where $\Delta ZPE$, $\Delta(\Delta U)$, and $\Delta S^{\Theta}$ respectively represent the variation of zero-point energy, internal energy variation, and entropy of the surface adsorbate for the adsorption process under the reaction condition. The thermodynamic corrections were calculated based on the partition functions from Boltzmann distribution. The detailed formulae could be found in our previous work[54–57]. $P$ and $p^{\Theta}$ are the saturated vapor pressure of adsorbate at 413.15 K and the standard vapor pressure, respectively. $x_i$ is the mole fraction of dissolved adsorbate in the liquid water. The negative $G_{ads}$ denotes that the adsorption process is thermodynamically favorable. The more negative $G_{ads}$, the stronger the adsorption.

The rectangle $p(6\times6)$ supercells of the three-layer exposed Pt(100) and $Pt_1Ni(100)$ surfaces were used to model the exposed facets of $0.2Pt/NiAl_2O_4$ and $0.2Pt5Ni/NiAl_2O_4$, respectively. For the $\gamma$-$Al_2O_3$ and $NiAl_2O_4$ supports, $1\times1$ surface supercells of the (111) facets were employed to represent the corresponding oxide surfaces. In all slab models, a vacuum region of 15 Å was introduced along the surface normal. Monkhorst-Pack k-point meshes of $2\times2\times1$ were used for the Pt(100) and $Pt_1Ni(100)$ slabs, while a $4\times4\times1$ meshes was adopted for both $\gamma$-$Al_2O_3$ (111) and $NiAl_2O_4$ (111). The geometry optimization was performed using the conjugate gradient algorithm until the maximal force of all the relaxed atoms lower than 0.05 eV/Å. The transition state was searched using the constrained optimization scheme[58]. and then verified by vibrational analysis, ensuring only one imaginary frequency. During the geometry optimization, the bottom layer metal atoms were fixed, while the upper two-layer surface atoms and adsorbates were allowed to relax.

The LOBSTER code was implemented to perform the Crystal Orbital Hamilton Population (COHP) analysis[59,60] which is capable of understanding the chemical bonding of BM molecules chemisorbed on the Pt(100) and $Pt_1Ni(100)$ surfaces[61]. The integration of the contribution of the energy bands of the projected crystal orbital Hamilton population up to the Fermi level (IpCOHP) indicates the bonding

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Author contribution line assigning DFT calculations.

catalysts, and performing the catalytic reactions. Q.X. and X.M.C.: DFT

Supplementary Figure S17 DFT adsorption energies/configurations.

became evident on both catalysts, with a more pronounced effect on the 2Pt catalyst.

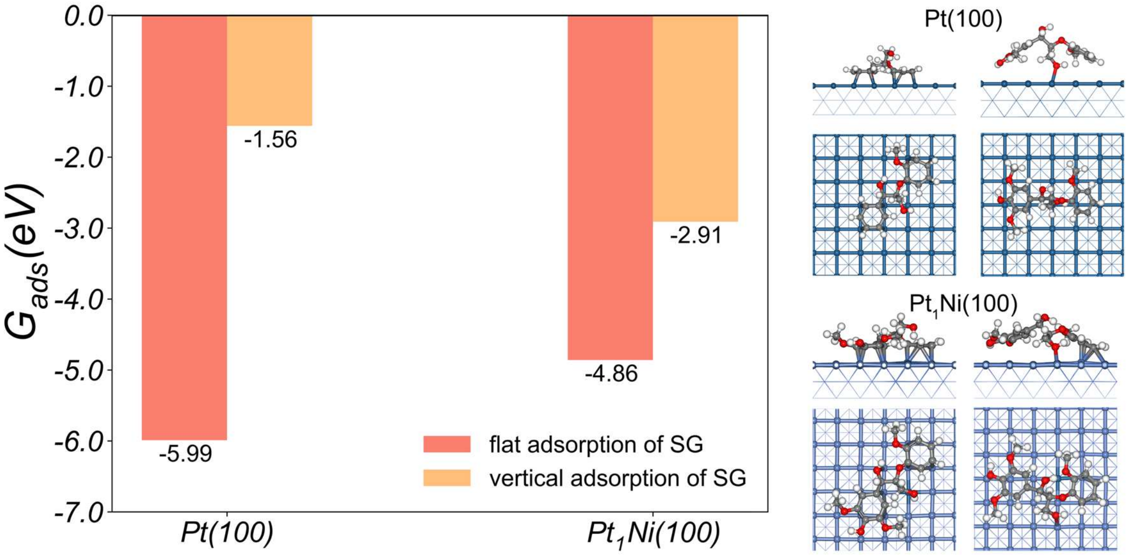


**Figure S17.** Free energies and adsorbed configurations of SG on Pt(100) and $Pt_1Ni(100)$.

Supplementary Figure S18 DFT free energy barriers.

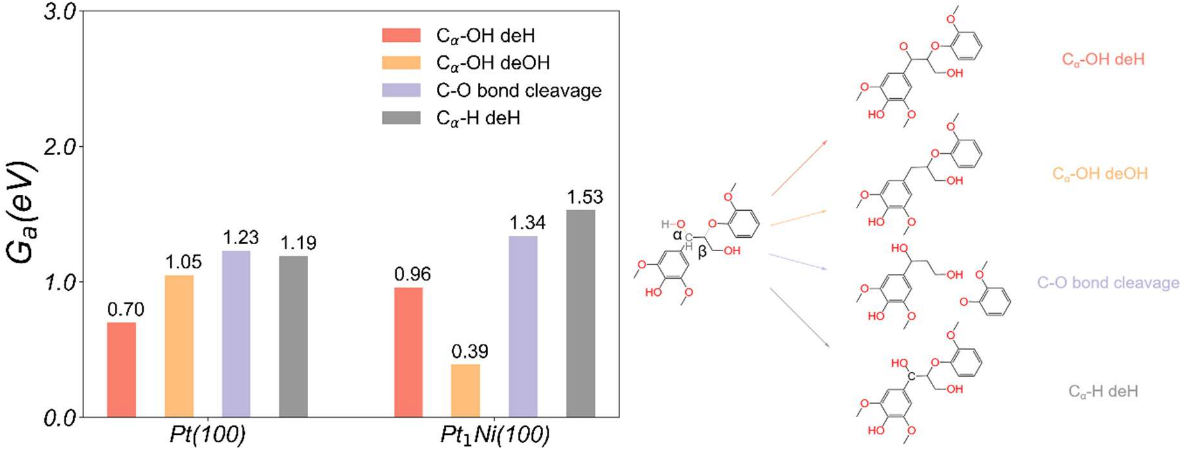


**Figure S18.** The free energy barriers of dehydrogenation of $C_\alpha$-OH, hydrogenolysis of $C_\alpha$-OH, C-O ether bond cleavage or dehydrogenation of $C_\alpha$-H of chemisorbed SG on Pt(100) and $Pt_1Ni(100)$.

Supplementary Figure S19 COHP analysis.

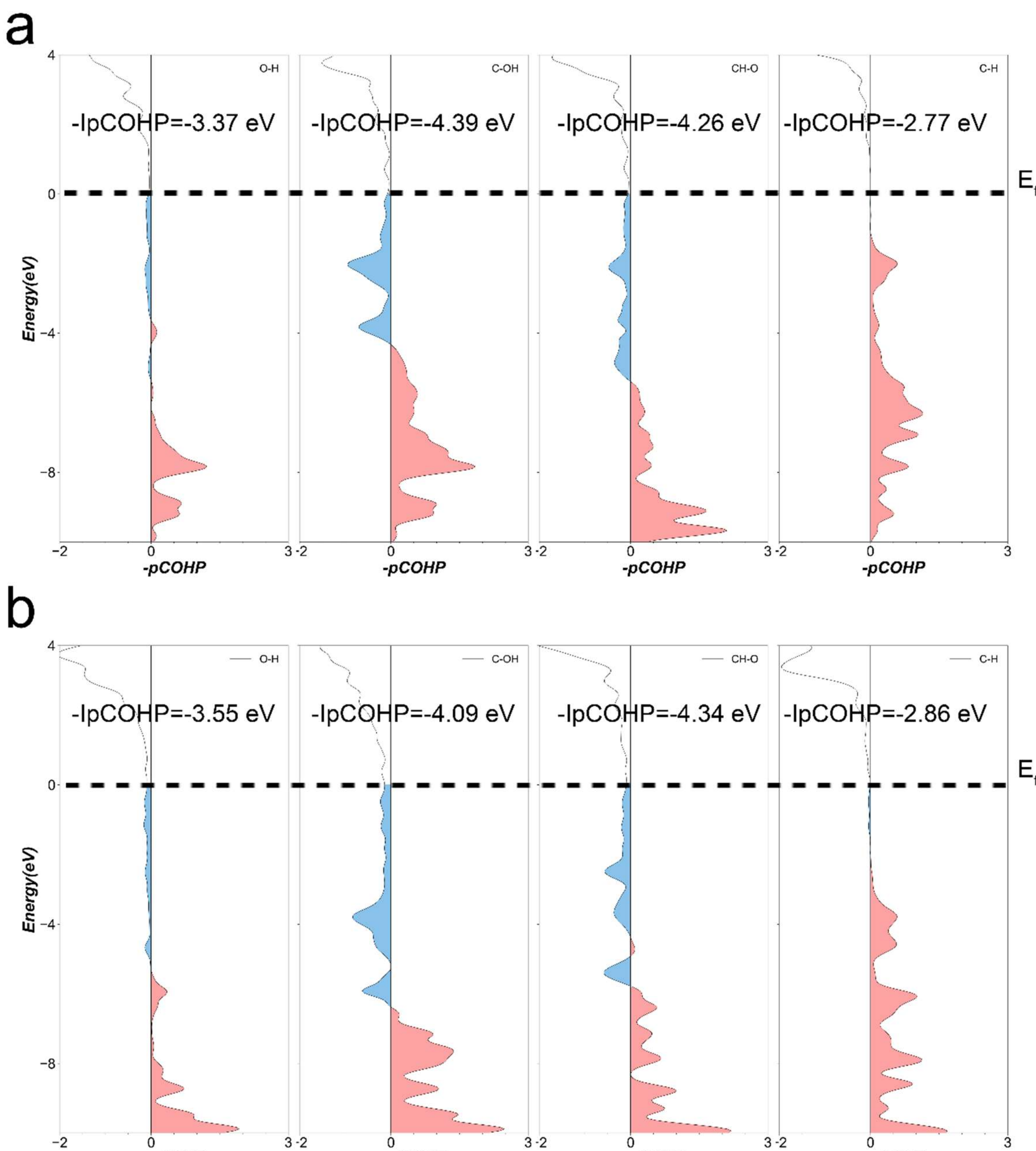


**Figure S19.** The COHP of the ($C_\alpha$)O-H, ($C_\alpha$)-OH, H$C_\beta$-O, and $C_\alpha$-H bonds of chemisorbed SG over the (a) Pt(100) and (b) $Pt_1Ni$(100) surfaces. (The red and blue areas respectively represent bonding and antibonding interactions)

Supplementary Figure S20 DFT pathway comparison.

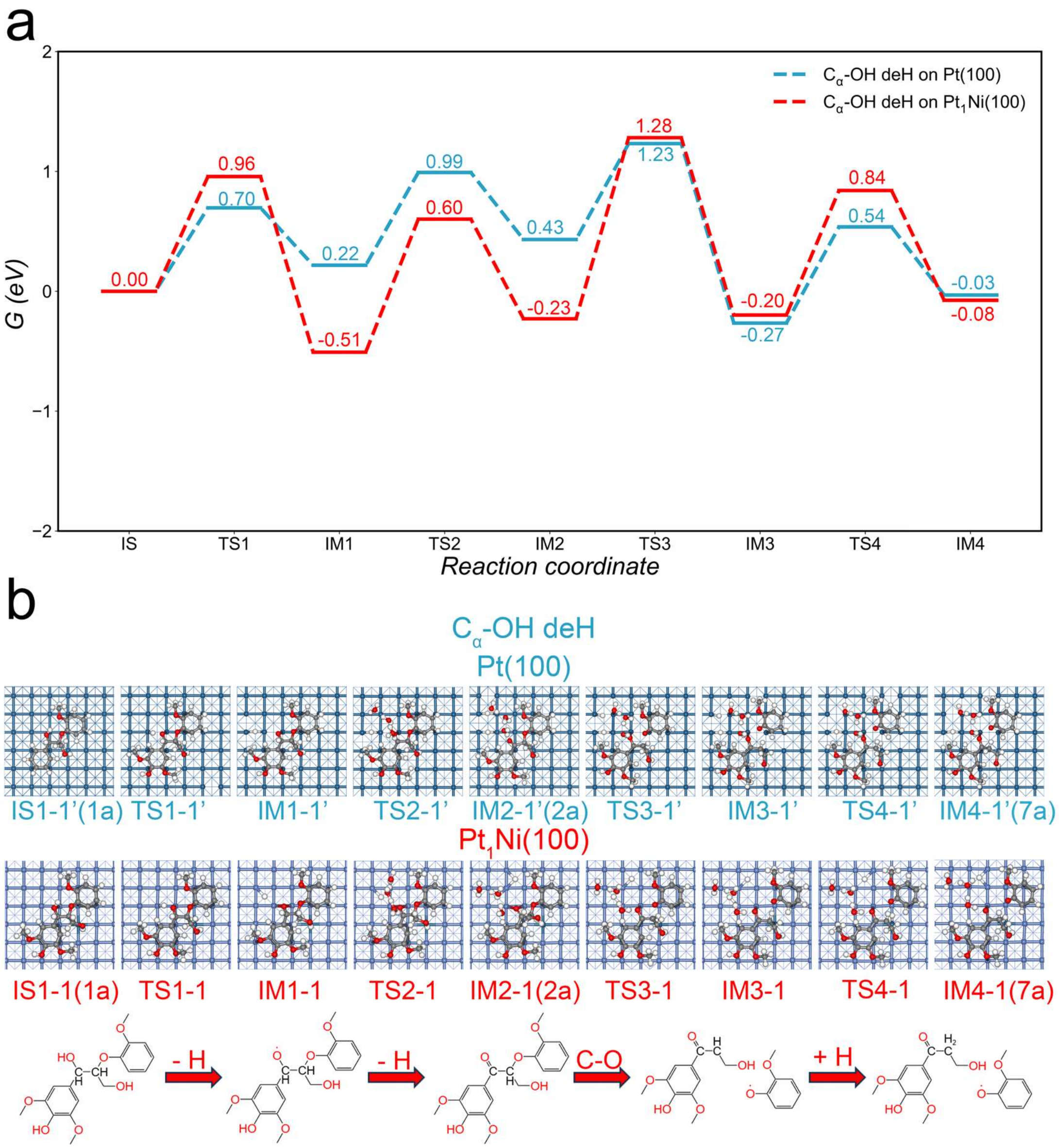


**Figure S20.** Comparison of the reaction pathways for the dehydrogenation of $C_α$-OH on Pt(100) and $Pt_1$Ni(100).

Supplementary Figure S21 DFT pathway comparison.

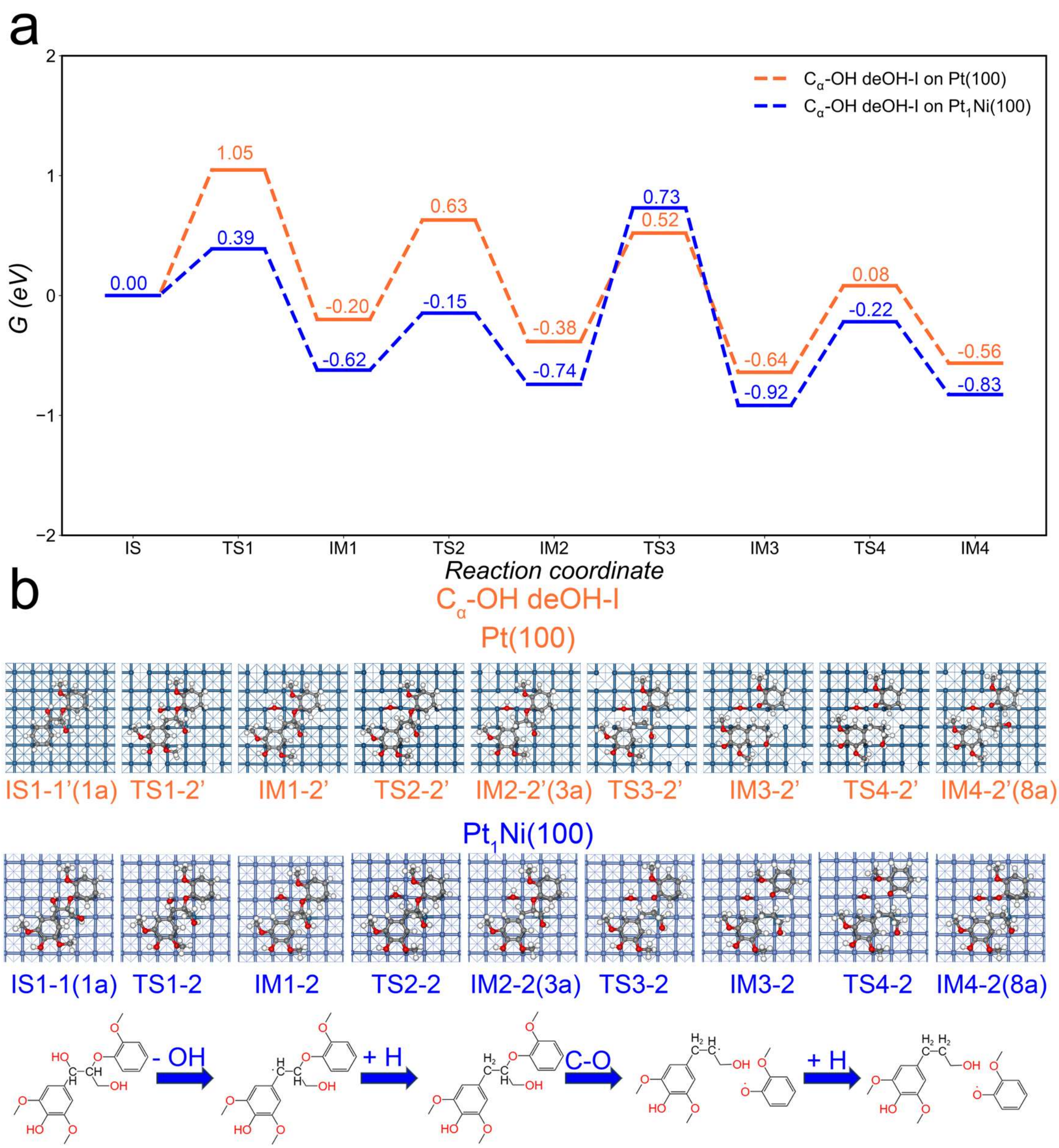


**Figure S21.** Comparison of the reaction pathways for the dehydroxylation of $C_\alpha$-OH I the on Pt(100) and $Pt_1Ni(100)$.

Supplementary Figure S22 DFT pathway comparison.

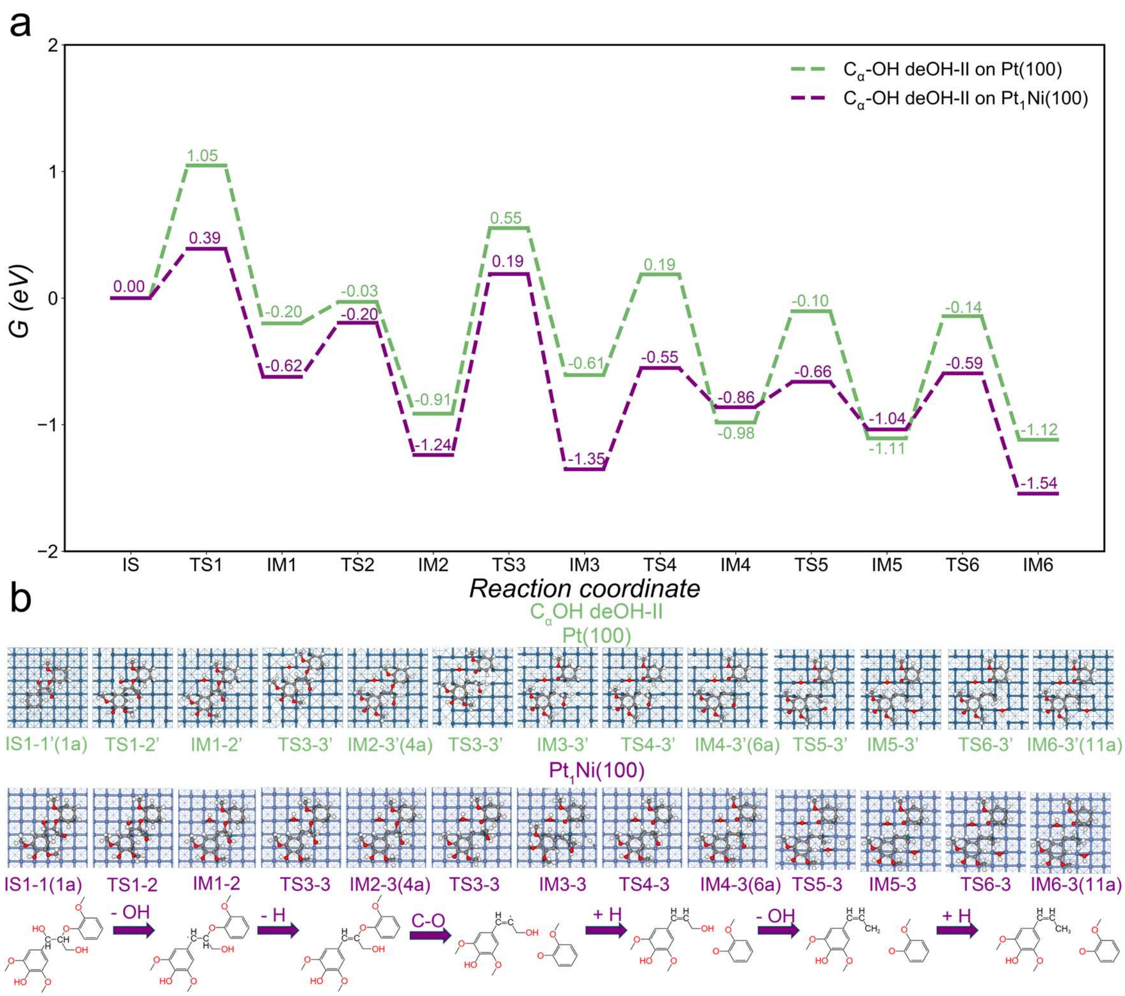


**Figure S22.** Comparison of the reaction pathways for the dehydroxylation of $C_\alpha$-OH II the on Pt(100) and $Pt_1Ni$(100).

Editorial/reviewer summary noting current DFT model problem.

support it with additional experimental evidence.

2. The current DFT modeling, using pristine Pt(100) to represent the “2Pt” single-atom catalyst and Pt1Ni(100) to represent the Pt1Ni material, is not an appropriate structural model for the catalysts used experimentally. The “2Pt” single-atom catalyst in this work is anchored on an oxide support (Pt–O environments) and therefore cannot be represented accurately by bulk Pt(100). Likewise, Ni in the synthesized catalyst is more likely present as NiO (or oxidized Ni species) rather than metallic Ni layers. Consequently, the present calculations risk drawing misleading mechanistic conclusions.

Other comments:

3. Reductive catalytic fractionation (RCF) is not equivalent to the lignin-first biorefining strategy. Please revise the introduction to clearly distinguish these two concepts and provide a more accurate and coherent summary of their

Reviewer #1 comments on DFT model, TS verification, and Figure 4 atom labels.

1. The authors attribute the enhanced Cα–OH activation by the oxophilicity of Ni to the higher selectivity towards valuable propenyl products. It was evidenced by the mechanistic studies over Pt1Ni(100) and Pt(100) based on DFT calculations. However, the specific role of the isolated Pt1 site in the Pt1Ni(100) model was not clearly explained. It should be clarified.

2. Self-hydrogen supplied fractionation rather than exogeneous hydrogen gas was used as the hydrogen resources in this work. Compared with direct use of hydrogen gas, whether would self-hydrogen supplied fractionation affect the reaction mechanism or the resultant selectivity?

3. Please clarify how the transition states were verified in the DFT calculations.

4. In the current version, the colors of the corresponding atoms in the figures of reaction mechanisms, such as Figure 4, were not labelled. It is difficult for readers to clearly identify the active sites and visualize the adsorption geometries, especially for the Pt1 in the Pt1Ni(100) surface.

Reviewer comment on inappropriate Pt/Pt1Ni slab model and start of reply.

2. The current DFT modeling, using pristine Pt(100) to represent the “2Pt” single-atom catalyst and Pt1Ni(100) to represent the Pt1Ni material, is not an appropriate structural model for the catalysts used experimentally. The “2Pt” single-atom catalyst in this work is anchored on an oxide support (Pt–O environments) and therefore cannot be represented accurately by bulk Pt(100). Likewise, Ni in the synthesized catalyst is more likely present as NiO (or oxidized Ni species) rather than metallic Ni layers. Consequently, the present calculations risk drawing misleading mechanistic conclusions.

Reply: We appreciate this important comment and agree that the correspondence between computational slab models and experimentally synthesized supported catalysts should be stated more explicitly to avoid confusion. In the revised manuscript, we have clarified what the DFT models represent. The reviewer's concern partly arises from an ambiguity in the description of the "2P" catalyst. In our experimental characterization, $2Pt/NiAl_2O_4$ is not a single-atom Pt catalyst. Instead, it contains Pt clusters (average size ~1.4 nm from HAADF-STEM shows in Figure S4) and shows a dominant linear CO-Pt band at ~2066 $cm^{-1}$ in Figure 2a, characteristic of Pt clusters. In addition, Pt-Pt coordination is observed by EXAFS fitting (Pt-Pt CN reported in Table S2). Therefore, Pt(100) was used as a simplified metallic Pt surface model to describe the intrinsic reactivity trend of Pt clusters, rather than a Pt-O single-atom site. We agree that Ni species on oxide supports can be oxidized prior to reduction. In our catalyst preparation, all samples were reduced in 10% $H_2$/Ar at 400°C for 3 h (see SI). Consistently, $H_2$-TPR shows that monometallic $5Ni/NiAl_2O_4$ exhibits a reduction peak attributable to NiO species, whereas Pt doping shifts the NiO reduction peaks to lower temperature, indicating that Pt promotes NiO reduction and facilitates the formation of metallic Ni species (Figure S2). More importantly, several *in-situ* and *ex-situ* characterizations support that the active $Pt_1Ni$ catalyst exposes a reduced Pt-Ni alloy surface with contiguous metallic Ni ensembles, rather than an oxidized NiO surface. *In-situ* CO-DRIFTS for 0.2Pt5Ni shows a band at 2034 $cm^{-1}$ (linear CO on isolated Pt atoms) together with a broad band at 1835 $cm^{-1}$ (bridged CO on contiguous Ni atoms), which is a characteristic signature of metallic Ni ensembles; in contrast, $5Ni/NiAl_2O_4$ exhibits almost no CO adsorption (Figure 2a). Pt $L_3$-edge XANES/EXAFS further indicate a more reduced Pt state in PtNi catalysts, and EXAFS fitting for 0.2Pt5Ni yields a dominant Pt-Ni coordination (CN ≈ 6.6) with no detectable Pt-Pt contribution, consistent with a SAA environment rather than Pt clusters or Pt-O anchored single sites (Figure 2b; Figure 2c; Table S2; Figure S5). Experiments show that $5Ni/NiAl_2O_4$ is inactive and its surface is primarily composed of NiO species, whereas $Pt_1Ni$ displays high activity, supporting that the $Pt_1Ni$ SAA effectively stabilizes metallic Ni under reaction-relevant conditions (Table 1). Accordingly, $Pt_1Ni(100)$ in DFT is intended to represent the local reduced Pt-Ni alloy surface (metallic Ni with isolated Pt atoms) that is evidenced experimentally, and is used to compare intrinsic mechanistic trends between a metallic Pt surface and a Pt-Ni SAA surface. We have added explicit statements in the revised manuscript to clearly distinguish these two surface models and to prevent potential misunderstanding.

towards valuable propenyl products. It was evidenced by the mechanistic studies over Pt1Ni(100) and Pt(100) based on DFT calculations. However, the specific role of the isolated Pt1 site in the Pt1Ni(100) model was

Author reply explaining the Pt1Ni(100) model and isolated Pt1 role.

Reply: We thank the reviewer for raising this comment. We agree that the specific role of the isolated $Pt_1$ site should be clarified more explicitly. Here we emphasize the role supported by experimental evidence: Pt stabilizes Ni in a metallic state rather than oxidized Ni species, thereby sustaining the active surface required for selective $C_\alpha$–OH activation. Experimentally, 0.2Pt5Ni forms a $Pt_1Ni$ single-atom alloy on Ni nanoparticles: atom-sized bright spots assigned to individual Pt atoms are observed on Ni NPs with no Pt clusters, while 2Pt5Ni shows mixed Pt single atoms and clusters. The *in-situ* CO-DRIFTS spectra further reveal a CO band at 2034 $cm^{-1}$ (linear CO on isolated Pt atoms) together with a broad band at 1835 $cm^{-1}$ assigned to bridged CO on contiguous Ni atoms, indicating the presence of metallic Ni ensembles adjacent to isolated Pt sites. In addition, Pt $L_3$-edge EXAFS fitting for 0.2Pt5Ni shows predominant Pt–Ni coordination (CN ≈ 6.6) and no Pt–Pt contribution, consistent with a single-atom alloy environment rather than Pt clusters. Crucially, $H_2$-TPR demonstrates that Pt doping makes NiO easier to reduce, shifting NiO reduction peaks to lower temperature relative to monometallic 5Ni, and 5Ni/$NiAl_2O_4$ is catalytically inactive with a surface primarily composed of NiO species. These observations collectively support that the $Pt_1Ni$ single-atom alloy structure promotes reduction and stabilizes metallic Ni species, which is essential for the observed high activity and selectivity in SCF. Therefore, in the $Pt_1Ni(100)$ model, the "$Pt_1$ site" represents the single atom Pt that alloy-stabilizes a metallic Ni surface. This stabilization maintains active metallic Ni ensembles that interact strongly with oxygen-containing functionalities, thereby favoring dehydroxylation pathways leading to valuable propenyl products.

Reviewer comment/reply about transition-state verification in DFT.

3. Please clarify how the transition states were verified in the DFT calculations.

Reply: Thanks for the comments, we have added explicit details in the revised SI (Computational details section) on transition-state (TS) verification. Briefly: Each TS structure was converged using constrained optimization scheme,[1] and then verified by vibrational analysis, ensuring only one imaginary frequency. These details have added to the supporting information.

1. Alavi, A.; Hu, P.; Deutsch, T.; Silvestrelli, P. L.; Hutter, J. CO oxidation on Pt (111): an ab initio density functional theory study. *Phys Rev Lett* **80**, 3650 (1998).

Reviewer comment/reply about Figure 4 color scheme and DFT adsorption geometries.

4. In the current version, the colors of the corresponding atoms in the figures of reaction mechanisms, such as Figure 4, were not labelled. It is difficult for readers to clearly identify the active sites and visualize the adsorption geometries, especially for the Pt1 in the Pt1Ni(100) surface.

Reply: Thanks for the comments, we have revised Figure 4 in below (Figure R8), by Adding a clear color legend for Pt, Ni, C, O, and H. Highlighting the isolated $Pt_1$ atom (with a red circle out). We have added

Figure R8 revised DFT comparative reaction pathway diagram.

these figures to the revised manuscript.

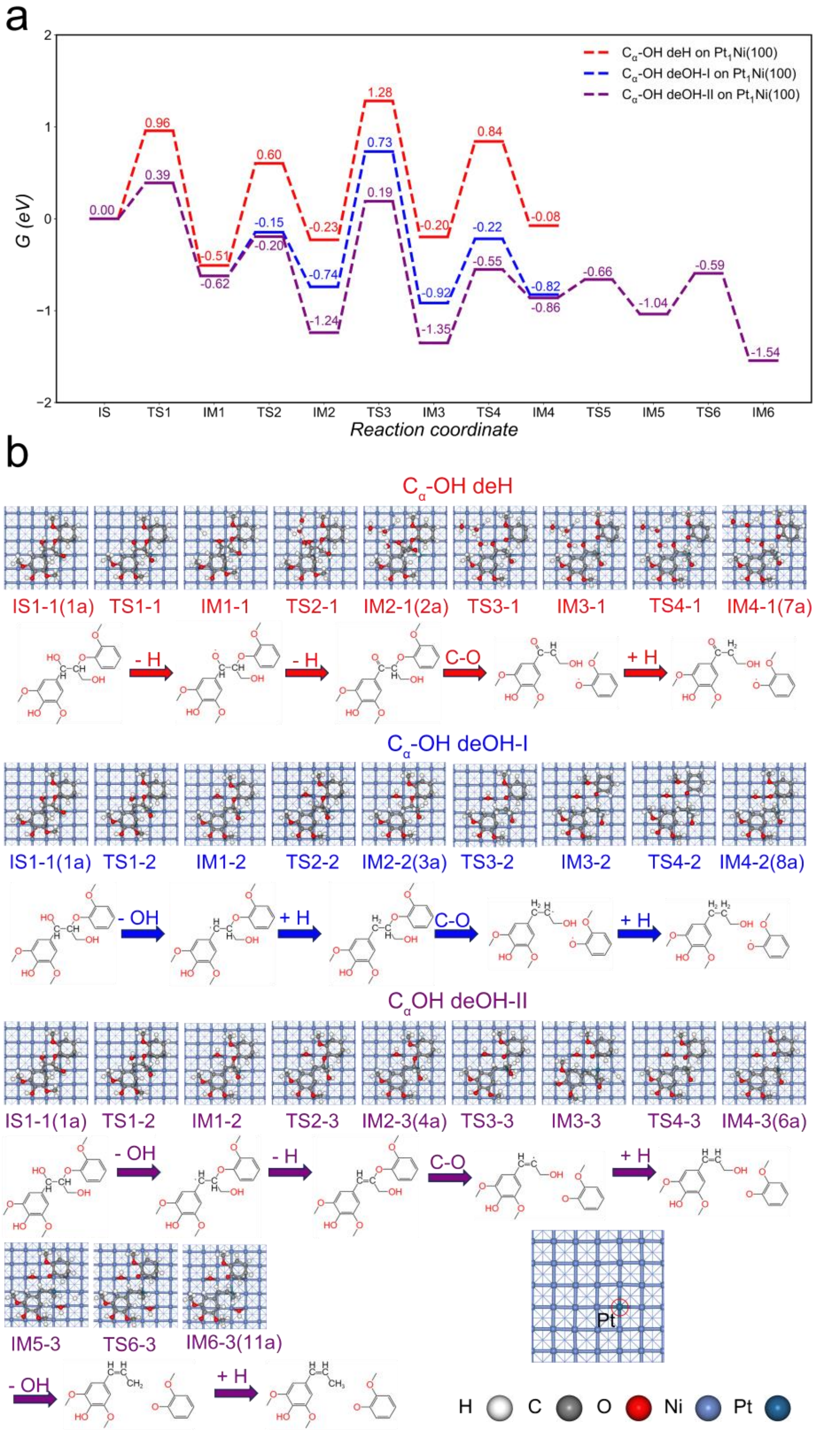


**Figure R8.** The comparative diagram of the reaction pathways. (a) energy profiles for $C_\alpha$-OH deH, $C_\alpha$-OH deOH-I, and $C_\alpha$-OH deOH-II on $Pt_1Ni(100)$. (b) corresponding the molecular structure diagram. The Pt atoms in $Pt_1Ni(100)$ have been circled in red in the last figure.

Second-round request that led to added support-level DFT calculations.

1) The manuscript suggests that $Al_2O_3$ and $NiAl_2O_4$ have different reaction pathways, which is driven by abundant oxygen vacancies, promoted hydrogen spillover, and robust metal-support interactions. The authors should further elaborate on the fundamental reasons underlying these observed differences.

Author reply reporting additional DFT calculations on Al2O3/NiAl2O4 supports.

**Reply:** We thank the reviewer for the insightful comment and have performed additional DFT calculations to elucidate the fundamental origins of the divergent reaction pathways observed on γ-$Al_2O_3$ and $NiAl_2O_4$ supports. As shown in Figure R1a, the reducible $NiAl_2O_4$ exhibits a substantially lower average oxygen vacancy formation energy (1.50 eV) compared to the non-reducible γ-$Al_2O_3$ (2.90 eV), indicating that $NiAl_2O_4$ sustains a higher concentration of oxygen vacancies under reaction conditions. These abundant oxygen vacancies are expected to enhance metal–support interactions. Furthermore, the critical elementary step for spillover-mediated support participation involves hydrogen transfer from an OH species to a neighboring lattice oxygen. Figure R1b presents the computed Gibbs free energy barriers for this step. Notably, $NiAl_2O_4$ displays a moderate barrier ($G_a$ = 1.06 eV) for this OH-to-O hydrogen transfer, whereas the same process on γ-$Al_2O_3$ is kinetically prohibitive ($G_a$ = 2.31 eV). Consequently, even if OH species form at the metal–support interface, γ-$Al_2O_3$ cannot effectively relay hydrogen migration across the support, limiting spillover and confining reactivity largely to the metal and its immediate vicinity. In contrast, the combination of higher reducibility and a much lower hydrogen-transfer barrier on $NiAl_2O_4$ facilitates rapid interfacial spillover, thereby promoting lattice-assisted hydrogenation pathways and shifting the dominant reaction mechanism.

Figure R1 oxygen-vacancy and hydrogen-transfer DFT calculations.

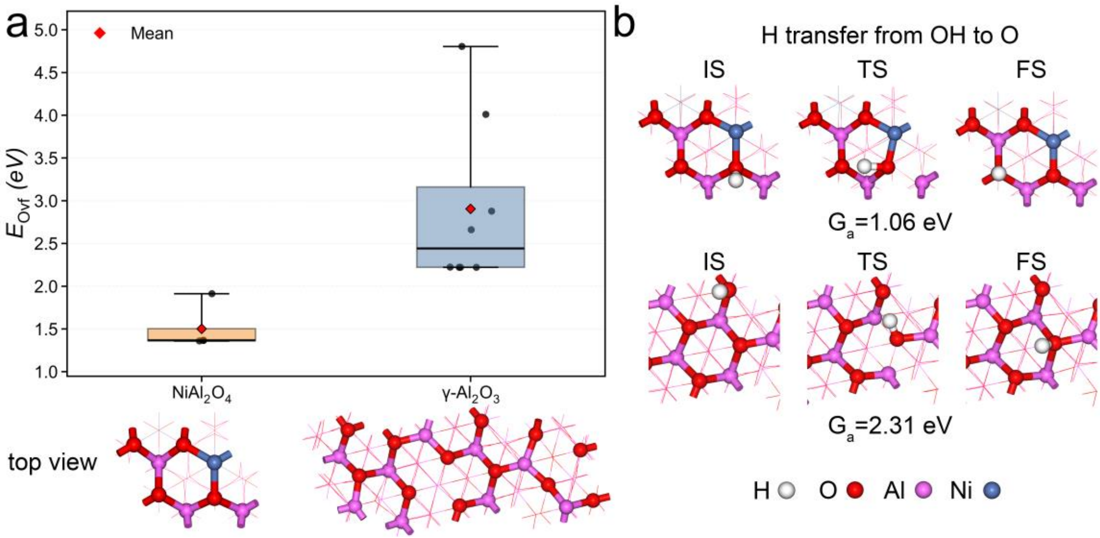


Figure R1. (a) Oxygen vacancy formation energies ($E_{Ovf}$) at different surface O sites on pristine $NiAl_2O_4$ (111) and γ-$Al_2O_3$ (111). (b) The Gibbs free energy barriers ($G_a$) for hydrogen transfer from an adsorbed hydroxyl (OH) to a neighboring lattice oxygen on $NiAl_2O_4$ (111) and γ-$Al_2O_3$ (111), together with the optimized initial state (IS), transition state (TS), and final state (FS) geometries. Color code: H, white; O, red; Al, purple; Ni, blue.